Critical Review: An Empirical Analysis of International Labour Migration in the Philippines

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Agbola, F. W. & Acupan, A. B. (2010)

Economic Systems, 34, 386–396

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The article at a glance

The study asks which push factors in the Philippines lead Filipinos to emigrate.

Why it matters

- **8.2 million** Filipinos live or work in more than 193 countries
- **US$14.4 billion** in remittances (money sent home) in 2007
- The Philippines has the **third-largest diaspora** in the world
- Emigration is now a **permanent part** of the economy

About the article

Authors: Frank W. Agbola and Angelito B. Acupan

Journal: *Economic Systems* (Elsevier), 2010

Data: yearly national data, 1975–2005 (31 years)

Method: error correction model (ECM)

Type: quantitative, time-series study

Key terms used in this review

Each term keeps the same meaning on every slide.

Term	Meaning
Emigration	People leaving the Philippines to live or work abroad
Migration share	Filipino migrants as a share of the population; what the study explains
Push / pull factor	A condition at home that pushes people out / a condition abroad that pulls them in
Proxy	A stand-in measure for something that cannot be measured directly
Stationary	Data that move around a stable average; checked with a unit root test
Cointegration	Data series that drift over time but still move together in the long run
ECM and ECT	Error correction model; its error correction term shows how fast data return to the long-run path
Statistically significant	Unlikely to be due to chance (here, at the 5% level)

1. Research problem and gap

Emigration became permanent, but its causes were "not well established."

The problem

- In the 1970s, sending workers abroad began as a **short-term fix** for unemployment
- It became a **permanent part** of the economy
- The law is **unclear**: RA 8042 says the State "does not promote" overseas work, but deployment "shall be encouraged"

The gap

- The causes of emigration from the Philippines are **not well established**
- Policymakers need **evidence** on which push factors matter most

Research question: How do economic, demographic and political factors affect emigration from the Philippines?

2. Theoretical framework

The authors combine four theories into one model of push factors.

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1. **Ravenstein (1885); Todaro (1969):** people migrate when they expect a better life abroad
2. **Human capital theory (Hicks 1939):** migration is an investment; more educated people migrate more
3. **Roy effect (Borjas 1987):** income inequality decides which skill groups leave
4. **Gravity model (Greenwood 1975):** migration depends on income, population and the cost of migration

The model in words

Migration share = *f*(cost of migration, adult literacy, income inequality, income inequality^2, political instability, inflation, population density, unemployment, per capita income)

Critical view: a mix of theories, not one model. Only push factors are used; pull factors from destination countries are left out.

3. Data

Yearly national data, 1975–2005: only 31 observations, all secondary data.

Variable	Measure / proxy	Source	Exp. sign
Migration share	Migrants (temporary + permanent) ÷ population	CFO; NSCB	Explained
Cost of migration	Average one-way airfare, Manila → top 24 destinations (proxy)	Authors' estimate	−
Adult literacy	Adult literacy rate (proxy for education)	BSP; NSCB	+
Income inequality, inequality^2	Gini coefficient and its square (Roy effect)	Not stated	±
Political instability	Number of leadership changes, coups and uprisings	Gastil (1987)	−
Inflation	Inflation rate (consumer prices)	IMF; BSP; NSCB	−
Population density	People per km^2	NSCB	+
Unemployment	Unemployment rate	IMF; BSP; NSCB	+
Per capita income	Real GDP per person	IMF; BSP; NSCB	−

Critical view: the source of the income inequality data is not stated, and the migration share mixes temporary and permanent migrants.

4. Methodology

A standard five-step time-series method, carefully applied.

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1. **Unit root tests:** ADF and P–P tests show the data become stationary after first differences
2. **Cointegration tests:** Engle–Granger, Johansen and Pesaran tests show the variables move together in the long run
3. **Choosing the model:** general-to-specific; drop terms that are not statistically significant
4. **Estimation:** the ECM is estimated by ordinary least squares (OLS) in EViews 6.0
5. **Diagnostic tests:** checks for problems in the model; all passed

Watch-out: the Johansen test finds 5–8 long-run relationships, but the ECM keeps only one ECT.

5. Key findings: the results

Four push factors are statistically significant; the model explains 63% of changes.

Explanatory variable	Coefficient	*t*-statistic	Result (5% level)
Unemployment	+0.236	4.629	**Significant**
Adult literacy	−0.659	−5.225	**Significant**
Political instability	−0.482	−3.444	**Significant**
Population density	−0.254	−2.756	**Significant**
Income inequality	−171.195	−1.594	Not significant
Income inequality^2	+165.927	1.323	Not significant
Inflation	−0.036	−1.327	Not significant
Cost of migration	−0.002	−0.429	Not significant
Per capita income	−0.0002	−0.293	Not significant

Model fit: adjusted $R^2 = 0.63$ · ECT = −0.776 (78% of any gap closes within a year) · F-statistic = 5.848 ($p = 0.000$)

Source: Agbola and Acupan (2010), Table 4.

5. Key findings: four significant push factors

Unemployment raises emigration; three other push factors lower it.

Unemployment ↑ → emigration ↑

Coefficient +0.236. Emigration is a “safety valve” when there are not enough jobs.

Adult literacy ↑ → emigration ↓

Coefficient −0.659. The opposite of human capital theory. The authors say most migrants do low-skilled jobs.

Political instability ↑ → emigration ↓

Coefficient −0.482. The migration share fell in unstable years (1983, 1986, 2001, 2005).

Population density ↑ → emigration ↓

Coefficient −0.254. The opposite of what was expected; the authors cannot fully explain it.

5. Key findings: other results and conclusions

Income, inflation, migration cost and inequality do not significantly affect emigration.

Not statistically significant

- **Cost of migration** and **per capita income**: negative, as expected
- **Inflation**: negative, as expected
- **Income inequality** and **income inequality^2**: the Roy effect does not hold for the Philippines

The authors' conclusions

- **Main push factors:** unemployment, adult literacy, political instability, population density
- **Collect and track** migration data better
- **Sign agreements** with destination countries to protect overseas Filipino workers
- **Build a home economy** with enough jobs, so the country relies less on emigration

Critical assessment of the study

Careful methods, but a small sample, weak proxies and missing pull factors.

Strengths

- **Important question** for a country that sends many workers abroad
- **Careful steps:** unit root tests, cointegration tests, then the ECM
- **All diagnostic tests** reported and passed
- **Clear policy background** (RA 8042)

Weaknesses

- **Small sample:** 31 yearly observations for 10 explanatory terms
- **Weak proxies:** airfare for cost of migration; adult literacy for education
- **Push factors only:** pull factors from destination countries are missing
- **Conflicting explanation:** “low-skilled migrants” clashes with data the paper cites on professionals leaving
- **Unclear tests:** per capita income is not stationary in the P–P test; no tests for structural breaks

6. Limitations and areas for further research

Better data and pull factors are the next steps.

Limitations the authors identify

- Migration data are **hard to collect** and compile
- Airfare shows only the **minimum cost** of migration
- The **population density** result is hard to explain
- The data **cannot prove** the political instability story

Future research (my suggestions)

1. **Split the data:** temporary vs. permanent migrants, and by skill level
2. **Add pull factors:** wages and job demand in destination countries
3. **Update the data:** go past 2005 and test structural breaks (1997, 2008, COVID-19)
4. **Use household data:** test human capital theory with survey data on migrants

TAKEAWAY

Push factors at home shape Filipino emigration.

- **Unemployment raises emigration;** adult literacy, political instability and population density lower it
- **Policy message:** build a home economy with enough jobs
- **Read with care:** small sample, weak proxies and missing pull factors

Thank you

Questions and discussion